

Lecture Notes: Fourier Analysis of Continuous-Time Signals

Course: 2110203 Computer Engineering Mathematics II

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1 Introduction

1.1 Motivation

Fourier analysis provides a systematic way of representing a signal as a linear combination of complex exponential functions. Once a signal is expressed in this form, its behaviour can be described in terms of the frequencies it contains rather than in terms of its variation in time. This alternative description is called the **frequency-domain representation** of the signal.

Two closely related tools are developed in this lecture:

- The **Fourier series**, which represents a *periodic* continuous-time signal as a sum of harmonically related complex exponentials. The resulting frequency-domain description is a *discrete* set of coefficients.
- The **Fourier transform**, which represents an *aperiodic* continuous-time signal as an integral of complex exponentials. The resulting frequency-domain description is a *continuous* function of frequency.

The Fourier transform can be obtained as a limiting case of the Fourier series, and for this reason the two tools are presented in that order.

1.2 Overview of Frequency-Domain Representations

Each class of signal has its own frequency-domain representation. Table 1 summarises the representations encountered in this course. The present lecture is concerned with the first two rows.

Signal class	Time domain	Frequency domain
Continuous-time, periodic	$x(t)$	a_n (Fourier series coefficients)
Continuous-time, aperiodic	$x(t)$	$X(j\omega)$ (Fourier transform)
Discrete-time, periodic	$x[n]$	a_n (discrete-time Fourier series)
Discrete-time, aperiodic	$x[n]$	$X(e^{j\omega})$ (discrete-time Fourier transform)
Discrete-time, finite length	$x[n]$	$X[k]$ (discrete Fourier transform)

Table 1: Time-domain and frequency-domain representations of the main signal classes.

1.3 Response of LTI Systems to Complex Exponentials

The reason complex exponentials are chosen as building blocks is that they are **eigenfunctions** of linear time-invariant (LTI) systems.

Consider an LTI system with impulse response $h(t)$ and let the input be the complex exponential

$$x(t) = e^{st},$$

where $s = \sigma + j\omega$ is a complex number. The output is obtained from the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)} d\tau = e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau)e^{-s\tau} d\tau}_{H(s)}.$$

Therefore

$$y(t) = H(s)e^{st},$$

where $H(s)$ is a complex constant that does not depend on t . The function e^{st} is the eigenfunction and $H(s)$ is the corresponding eigenvalue. In words, the response of an LTI system to a complex exponential is the *same* complex exponential, changed only in amplitude and phase.

By linearity, if the input is a linear combination of complex exponentials,

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t},$$

then the output is obtained by scaling each component by its own eigenvalue:

$$y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}.$$

When s is restricted to the imaginary axis, $s = j\omega$, the eigenvalue $H(j\omega)$ is called the **frequency response** of the system. This observation is the practical justification for Fourier analysis: if a signal can be decomposed into complex exponentials, the response of an LTI system is obtained by multiplying each component by a single complex number.

2 Fourier Series Representation of Periodic Signals

The Fourier series is named after the French mathematician and physicist Jean Baptiste Joseph Fourier (1768–1830), who proposed that a periodic signal can be represented by a sum of sinusoids. Typical applications include the analysis of acoustic and electromagnetic waves, the vertical displacement of a mechanical pendulum, the periodic vibrations of musical instruments, and the periodic structure of crystals.

2.1 Periodic Signals and Harmonically Related Complex Exponentials

A continuous-time signal $x(t)$ is periodic if, for some positive non-zero value of T ,

$$x(t) = x(t + nT), \quad n = 1, 2, 3, \dots$$

The smallest such positive value of T is the **fundamental period**, and

$$\omega_0 = \frac{2\pi}{T}$$

is the **fundamental radian frequency**, measured in radians per second.

Both the sinusoidal signal $x(t) = \cos(\omega_0 t)$ and the complex exponential $x(t) = e^{j\omega_0 t}$ are periodic with fundamental frequency ω_0 and fundamental period $T = 2\pi/\omega_0$.

The set of **harmonically related complex exponentials** is defined as

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \dots$$

Every member of this set is periodic with period T ; the fundamental period of $\phi_k(t)$ is $T/|k|$ for $k \neq 0$. The set is **orthogonal** over any interval of length T :

$$\int_{\langle T \rangle} \phi_k(t) \phi_l^*(t) dt = \int_{\langle T \rangle} e^{j(k-l)\omega_0 t} dt = \begin{cases} T, & k = l, \\ 0, & k \neq l, \end{cases}$$

where $\int_{\langle T \rangle}$ denotes integration over any interval whose length is one period. Orthogonality is the property that makes the Fourier coefficients easy to compute, as shown in Section 2.4.

2.2 The Synthesis Equation

A linear combination of harmonically related complex exponentials,

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{jn(2\pi/T)t},$$

is called the **Fourier series representation** of $x(t)$, and the complex numbers a_n are the **Fourier series coefficients**, also called the spectral coefficients. This expression is the **synthesis equation**: it builds the signal from its frequency components.

The individual terms are interpreted as follows:

- $n = 0$: the term a_0 is a constant. It is the DC component, equal to the average value of the signal over one period.
- $n = 1, -1$: the fundamental components, or first harmonic components. They oscillate at the fundamental frequency ω_0 .
- $n = 2, -2$: the second harmonic components. They oscillate at twice the fundamental frequency and have half the fundamental period.
- $n = N, -N$: the N th harmonic components, oscillating at $N\omega_0$.

2.3 Alternative Forms for Real-Valued Signals

If $x(t)$ is real, then $x^*(t) = x(t)$, and taking the conjugate of the synthesis equation gives

$$x(t) = \sum_{n=-\infty}^{\infty} a_n^* e^{-jn\omega_0 t}.$$

Replacing the summation index n by $-n$ yields

$$x(t) = \sum_{n=-\infty}^{\infty} a_{-n}^* e^{jn\omega_0 t}.$$

Comparing this expression with the synthesis equation gives the **conjugate symmetry** property

$$a_n = a_{-n}^*, \quad \text{equivalently} \quad a_n^* = a_{-n}.$$

Using this property, the summation can be folded onto the positive harmonics only:

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n e^{jn\omega_0 t} + a_{-n} e^{-jn\omega_0 t}] = a_0 + \sum_{n=1}^{\infty} [a_n e^{jn\omega_0 t} + a_n^* e^{-jn\omega_0 t}].$$

Since the two terms inside the summation are complex conjugates of each other, their sum is twice the real part of either one:

$$x(t) = a_0 + \sum_{n=1}^{\infty} 2 \operatorname{Re} \{a_n e^{jn\omega_0 t}\}.$$

Two further forms follow. Writing the coefficients in rectangular form as $a_n = B_n + jC_n$, where B_n and C_n are both real, gives the **trigonometric form**

$$x(t) = a_0 + 2 \sum_{n=1}^{\infty} [B_n \cos(n\omega_0 t) - C_n \sin(n\omega_0 t)].$$

Writing them in polar form as $a_n = |a_n| e^{j\theta_n}$ gives the **amplitude-phase form**

$$x(t) = a_0 + 2 \sum_{n=1}^{\infty} |a_n| \cos(n\omega_0 t + \theta_n).$$

All three forms describe the same signal; the choice is a matter of convenience.

2.4 Determination of the Fourier Coefficients

Assume that a given periodic signal can be represented by the synthesis equation, with the summation index renamed to k so that it is not confused with the particular harmonic to be extracted. To obtain the coefficient of harmonic n , multiply both sides by $e^{-jn\omega_0 t}$:

$$x(t)e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t}.$$

Integrating both sides over one period, from 0 to $T = 2\pi/\omega_0$, and interchanging the order of integration and summation gives

$$\int_0^T x(t)e^{-jn\omega_0 t} dt = \sum_{k=-\infty}^{\infty} a_k \left[\int_0^T e^{jk\omega_0 t} e^{-jn\omega_0 t} dt \right].$$

The bracketed integral is evaluated with Euler's relation. For $k \neq n$,

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \left. \frac{e^{j(k-n)\omega_0 t}}{j(k-n)\omega_0} \right|_{t=0}^T = \frac{e^{j(k-n)\omega_0 T} - 1}{j(k-n)\omega_0} = 0,$$

because $\omega_0 T = 2\pi$ and therefore $e^{j2\pi(k-n)} = 1$. For $k = n$ the integrand equals unity and the integral equals T . In summary,

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \begin{cases} T, & k = n, \\ 0, & k \neq n. \end{cases}$$

Only the single term $k = n$ survives on the right-hand side, so that

$$\int_0^T x(t)e^{-jn\omega_0 t} dt = a_n T,$$

which gives the **analysis equation**

$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t)e^{-jn\omega_0 t} dt.$$

2.5 Summary of the Fourier Series Pair

The set of complex exponentials

$$\phi_n(t) = e^{jn(2\pi/T)t}$$

forms an orthogonal set, and a periodic signal $x(t)$ is represented as a sum of these basis functions. The Fourier series pair is

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{jn(2\pi/T)t} \quad (\text{synthesis}),$$

$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t)e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t)e^{-jn(2\pi/T)t} dt \quad (\text{analysis}).$$

2.6 Line Spectra

The coefficients a_n of the exponential Fourier series define a complex-valued function of the discrete frequency $n\omega_0$. Their properties are summarised below.

- Each frequency present in the signal is an integer multiple of the fundamental frequency; in hertz, the harmonics are located at multiples of $\omega_0/2\pi$.
- The DC component is equal to a_0 and is obtained by setting $n = 0$.
- The plot of $|a_n|$ versus $n\omega_0$ displays the amplitude of the various frequency components and is called the **magnitude spectrum**.

- The plot of $\angle a_n$ versus $n\omega_0$ is called the **phase spectrum** of $x(t)$.
- Taken together, the plots of $|a_n|$ and $\angle a_n$ are called the **line spectra** of the signal, because the spectrum exists only at discrete frequencies and is therefore drawn as a set of lines.
- For real-valued signals, conjugate symmetry implies that the magnitude spectrum is an even function and the phase spectrum is an odd function of n :

$$|a_n| = |a_{-n}| \quad \text{and} \quad \angle a_n = -\angle a_{-n}.$$

2.7 Worked Example: Line Spectra of a Square Wave

Find the line spectra of the periodic signal

$$x(t) = \begin{cases} -K, & -1 < t < 0, \\ K, & 0 < t < 1, \end{cases} \quad x(t+2) = x(t).$$

The fundamental period is $T = 2$, hence $\omega_0 = 2\pi/2 = \pi$.

From the analysis equation,

$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn(2\pi/T)t} dt = \frac{1}{2} \int_{-1}^1 x(t) e^{-jn\pi t} dt = \frac{1}{2} \left[\int_{-1}^0 (-K) e^{-jn\pi t} dt + \int_0^1 K e^{-jn\pi t} dt \right].$$

Evaluating the two integrals for $n \neq 0$ gives

$$a_n = \frac{K}{jn\pi} \left[1 - \frac{1}{2} (e^{jn\pi} + e^{-jn\pi}) \right] = \frac{K}{jn\pi} [1 - \cos(n\pi)],$$

so that

$$a_n = \begin{cases} \frac{2K}{jn\pi}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

The case $n = 0$ must be treated separately, because the expression above is indeterminate at $n = 0$. Direct evaluation gives

$$a_0 = \frac{1}{2} \int_{-1}^1 x(t) dt = 0,$$

as expected, since the signal has zero average value.

Writing $2K/(jn\pi) = -j2K/(n\pi)$ shows that the coefficients are purely imaginary. The line spectra are therefore

$$|a_n| = \frac{2K}{|n|\pi} \quad (n \text{ odd}), \quad \angle a_n = \begin{cases} -\pi/2, & n > 0 \text{ odd}, \\ +\pi/2, & n < 0 \text{ odd}, \end{cases}$$

and $a_n = 0$ for even n . The magnitude spectrum is even and the phase spectrum is odd, in agreement with the fact that $x(t)$ is real.

Substituting the coefficients into the synthesis equation and combining conjugate pairs gives the trigonometric form

$$x(t) = \frac{4K}{\pi} \sum_{n \text{ odd}, n \geq 1} \frac{\sin(n\pi t)}{n}.$$

Two observations are worth noting. First, only odd harmonics are present, which is a consequence of the half-wave symmetry of this particular waveform. Second, the coefficients decay only as $1/n$, which is a slow rate of decay caused by the discontinuities in $x(t)$: a signal with jumps requires many high-frequency components.

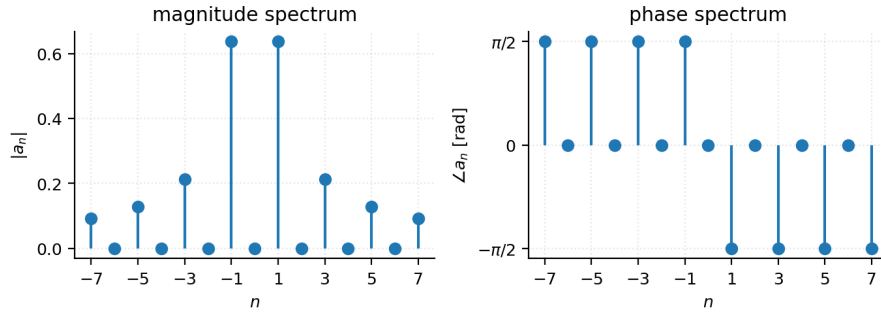


Figure 1: Line spectra of the square wave with $K = 1$. The magnitude spectrum $|a_n|$ is an even function of n and decays as $1/|n|$, while the phase spectrum $\angle a_n$ is odd and takes the values $\mp\pi/2$. Even-indexed coefficients are zero.

2.8 Partial Sums and the Gibbs Phenomenon

Whether the synthesis equation actually reproduces the signal it was built from is a question of convergence. A sufficient set of conditions is the **Dirichlet conditions** for the Fourier series, which require that, over one period,

1. $x(t)$ is absolutely integrable, that is $\int_{\langle T \rangle} |x(t)| dt < \infty$;
2. $x(t)$ has a finite number of minima and maxima;
3. $x(t)$ has a finite number of discontinuities, and each of them is finite.

These conditions are sufficient but not necessary, and every signal encountered in this lecture satisfies them. The analogous conditions for the Fourier transform are stated in Section 3.3.

The truncated Fourier series, or partial sum,

$$x_N(t) = \sum_{n=-N}^N a_n e^{jn\omega_0 t},$$

is an approximation of $x(t)$ obtained by keeping the harmonics up to order N . When the Dirichlet conditions hold, the approximation improves as N increases and converges to $x(t)$ at every point where the signal is continuous. At a discontinuity, the partial sum converges to the midpoint of the jump.

Near a discontinuity the partial sum overshoots the signal, and the peak overshoot does not vanish as N increases; instead, the ripples become narrower and move closer to the discontinuity. This behaviour is known as the **Gibbs phenomenon** and is illustrated in Figure 2.

2.9 Worked Example: Line Spectra of a Sum of Sinusoids

Find the line spectra of the periodic signal

$$x(t) = 1 + \sin(\omega_0 t) + 2 \cos(\omega_0 t) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right).$$

When a signal is already given as a sum of sinusoids, the coefficients are obtained by inspection after applying Euler's relations

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}, \quad \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}.$$

Substituting these expressions gives

$$x(t) = 1 + \frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t} + e^{j\omega_0 t} + e^{-j\omega_0 t} + \frac{1}{2}e^{j\pi/4}e^{j2\omega_0 t} + \frac{1}{2}e^{-j\pi/4}e^{-j2\omega_0 t}.$$

Collecting the terms that multiply each $e^{jn\omega_0 t}$, and using $1/(2j) = -j/2$, gives

$$a_0 = 1, \quad a_1 = 1 - \frac{j}{2}, \quad a_{-1} = 1 + \frac{j}{2}, \quad a_2 = \frac{1}{2}e^{j\pi/4}, \quad a_{-2} = \frac{1}{2}e^{-j\pi/4},$$

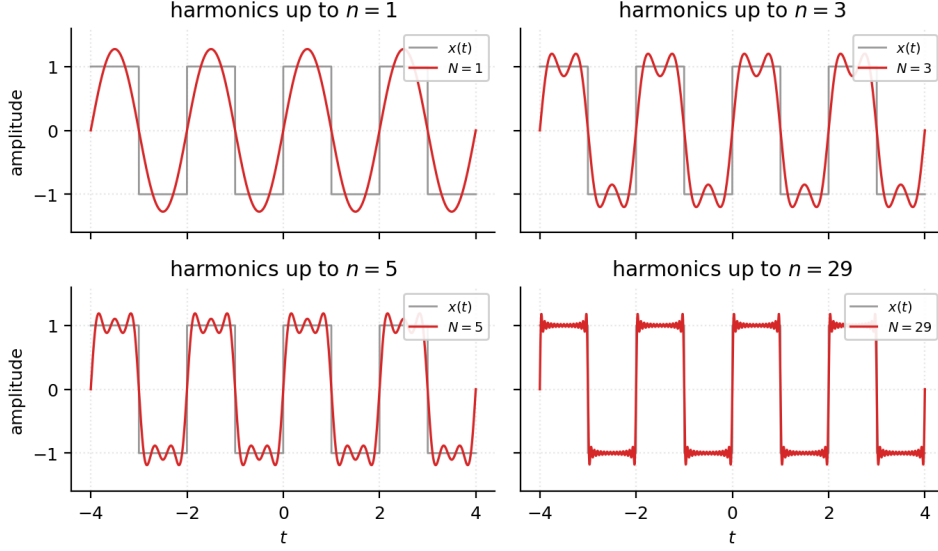


Figure 2: Partial sums of the Fourier series of the square wave of Section 2.7 with $K = 1$. The grey curve is the original signal and the coloured curve is the partial sum obtained by keeping harmonics up to order N . The overshoot near each discontinuity does not disappear as N increases; the ripples merely become narrower.

and $a_n = 0$ for $|n| \geq 3$. The magnitudes and phases follow from

$$|a_n| = \sqrt{(\operatorname{Re}\{a_n\})^2 + (\operatorname{Im}\{a_n\})^2}, \quad \angle a_n = \tan^{-1} \left(\frac{\operatorname{Im}\{a_n\}}{\operatorname{Re}\{a_n\}} \right),$$

and are collected in Table 2.

n	a_n	$ a_n $	$\angle a_n$
0	1	1	0
1	$1 - j/2$	$\sqrt{5}/2 \approx 1.118$	$-\tan^{-1}(1/2) \approx -0.464$ rad
-1	$1 + j/2$	$\sqrt{5}/2 \approx 1.118$	$+\tan^{-1}(1/2) \approx +0.464$ rad
2	$\frac{1}{2}e^{j\pi/4} = \frac{\sqrt{2}}{4}(1 + j)$	$1/2$	$+\pi/4$
-2	$\frac{1}{2}e^{-j\pi/4} = \frac{\sqrt{2}}{4}(1 - j)$	$1/2$	$-\pi/4$

Table 2: Fourier series coefficients of $x(t) = 1 + \sin(\omega_0 t) + 2 \cos(\omega_0 t) + \cos(2\omega_0 t + \pi/4)$. The coefficients satisfy $a_n = a_{-n}^*$, as required for a real-valued signal.

The phase of the fundamental component is exactly $-\tan^{-1}(1/2) \approx -0.4636$ rad, or about -26.57° . This value is not a rational multiple of π and should be left in the inverse-tangent form; in particular it is not $-\pi/6$.

2.10 Parseval's Relation for Periodic Signals

A periodic signal has infinite energy but finite average power. Parseval's relation states that this power may be computed either in the time domain or in the frequency domain:

$$P = \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |a_n|^2.$$

The result follows by substituting the synthesis equation into the left-hand side and using the orthogonality of the harmonically related exponentials. Its interpretation is that $|a_n|^2$ is the average power contributed by the n th harmonic, so the total average power is the sum of the powers of the individual harmonic components. For a real signal the terms n and $-n$ contribute equally, and the power in the n th harmonic pair is $2|a_n|^2$.

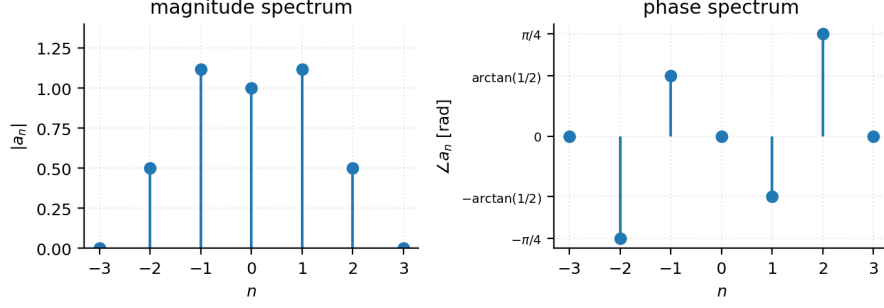


Figure 3: Line spectra of $x(t) = 1 + \sin(\omega_0 t) + 2 \cos(\omega_0 t) + \cos(2\omega_0 t + \pi/4)$. Only the DC term and the first two harmonics are present. The magnitude spectrum is even and the phase spectrum is odd.

As an illustration, the square wave of Section 2.7 has $P = K^2$ by direct time-domain calculation, and the coefficients give

$$\sum_{n \text{ odd}} \left(\frac{2K}{|n|\pi} \right)^2 = \frac{8K^2}{\pi^2} \sum_{n \text{ odd}, n \geq 1} \frac{1}{n^2} = \frac{8K^2}{\pi^2} \cdot \frac{\pi^2}{8} = K^2,$$

in agreement.

3 The Continuous-Time Fourier Transform

The Fourier series applies only to periodic signals. The continuous-time Fourier transform extends frequency-domain analysis to aperiodic signals such as $u(t)$, $e^{-t}u(t)$, and $\text{rect}(t/T)$, decomposing them into a weighted combination of complex exponentials. Together with the Fourier series, it provides the tools for the analysis of LTI systems.

The key idea is that an **aperiodic signal can be viewed as a periodic signal with an infinite period**. In the Fourier series representation of a periodic signal, as the period increases the fundamental frequency decreases and the harmonically related components become closer in frequency. As the period becomes infinite, the frequency components form a continuum and the Fourier series sum becomes an integral.

3.1 From the Fourier Series to the Fourier Transform

Let $x(t)$ be of finite duration, so that it is zero outside some finite interval, and let $\tilde{x}(t)$ be the periodic signal with period T that coincides with $x(t)$ over the interval $-T/2 < t < T/2$ and repeats outside it, with T large enough to contain the whole of $x(t)$. As T is increased, the neighbouring replicas move away from the origin, and in the limit $T \rightarrow \infty$ only a single pulse remains, so that $\tilde{x}(t) \rightarrow x(t)$ for every t . The finite-duration assumption keeps the argument simple; the resulting transform pair applies more generally.

The Fourier representation of $\tilde{x}(t)$ is

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}, \quad \text{where} \quad a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt.$$

In the limit as $T \rightarrow \infty$, the spacing between adjacent harmonics, $\omega_0 = 2\pi/T$, becomes an infinitesimally small quantity $d\omega$, so that

$$\frac{1}{T} \rightarrow \frac{d\omega}{2\pi},$$

and the discrete frequency $n\omega_0$ becomes a continuous variable ω . The limits of integration extend to $\pm\infty$. The Fourier coefficient per unit frequency interval therefore becomes

$$a_n = \frac{d\omega}{2\pi} \int_{t=-\infty}^{\infty} x(t) e^{-j\omega t} dt.$$

Substituting this result into the synthesis equation, the limit of the sum becomes an integral over ω and $\tilde{x}(t)$ approaches $x(t)$:

$$x(t) = \int_{\omega=-\infty}^{\infty} \left[\int_{\lambda=-\infty}^{\infty} x(\lambda) e^{-j\omega\lambda} d\lambda \right] e^{j\omega t} \frac{d\omega}{2\pi}.$$

A separate symbol λ has been used for the variable of integration in the inner integral to emphasise that this integral is a function of ω only, not of t . Denoting the inner integral by $X(j\omega)$ leads directly to the transform pair of the next section.

3.2 The Fourier Transform Pair

For an aperiodic signal $x(t)$, the **Fourier transform pair** is

$$\begin{aligned} X(j\omega) &= \int_{t=-\infty}^{\infty} x(t) e^{-j\omega t} dt && \text{(analysis, or forward transform),} \\ x(t) &= \frac{1}{2\pi} \int_{\omega=-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega && \text{(synthesis, or inverse transform).} \end{aligned}$$

There is a one-to-one correspondence between a signal and its transform, written

$$x(t) \longleftrightarrow X(j\omega),$$

and the notation $x(t) \leftrightarrow X(\omega)$ is also used. The transform is often denoted by the operator $\mathcal{F}$:

$$\mathcal{F}\{x(t)\} = X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt, \quad \mathcal{F}^{-1}\{X(j\omega)\} = x(t).$$

Since $X(j\omega)$ is in general complex, it is usually described by its magnitude and phase,

$$X(j\omega) = |X(j\omega)| e^{j\angle X(j\omega)},$$

where $|X(j\omega)|$ is the **magnitude spectrum** and $\angle X(j\omega)$ is the **phase spectrum**. Unlike the line spectra of a periodic signal, both are continuous functions of ω .

Two special values are often useful:

$$X(j0) = \int_{-\infty}^{\infty} x(t) dt, \quad x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega.$$

That is, the transform evaluated at $\omega = 0$ is the area under the signal, and the signal at $t = 0$ is proportional to the area under the transform.

3.3 Existence of the Fourier Transform

One set of conditions that guarantees that a signal has a Fourier transform is the **Dirichlet conditions**, which require that

1. $x(t)$ is absolutely integrable; that is,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty;$$

2. $x(t)$ has a finite number of minima and maxima within any finite interval;
3. $x(t)$ has a finite number of discontinuities within any finite interval, and each of these discontinuities is finite.

These conditions are sufficient but not necessary. Many signals of practical interest, such as the unit step $u(t)$, periodic signals, and the sampling function, are not absolutely integrable and yet possess Fourier transforms when impulses are allowed in the frequency domain. Such transforms are discussed in Section 5.

3.4 Worked Example: Transform of a Rectangular Pulse

Find the continuous-time Fourier transform of $x(t) = \text{rect}(t/\tau)$, the rectangular pulse of unit amplitude extending from $-\tau/2$ to $\tau/2$.

From the analysis equation,

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_{-\tau/2}^{\tau/2} (1)e^{-j\omega t} dt = \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{t=-\tau/2}^{\tau/2} = \frac{e^{-j\omega(\tau/2)} - e^{j\omega(\tau/2)}}{-j\omega}.$$

Using $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$, this reduces to

$$X(j\omega) = \frac{2 \sin(\omega\tau/2)}{\omega} = \frac{\tau \sin(\omega\tau/2)}{(\omega\tau/2)}.$$

With the sampling function defined as in Lecture 01,

$$\text{sinc}(x) = \frac{\sin x}{x}, \quad \text{sinc}(0) = 1,$$

the result is written compactly as

$$\boxed{\text{rect}(t/\tau) \longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)}.$$

Note that this course uses the unnormalised convention $\text{sinc}(x) = \sin(x)/x$; some texts define $\text{sinc}(x) = \sin(\pi x)/(\pi x)$ instead, which changes the location of the zero crossings by a factor of π .

The transform has the following features:

- $X(j0) = \tau$, which equals the area of the pulse.
- $X(j\omega)$ is real and even, as expected because $x(t)$ is real and even.
- The zero crossings occur where $\sin(\omega\tau/2) = 0$ with $\omega \neq 0$, that is at $\omega = 2\pi n/\tau$ for $n = \pm 1, \pm 2, \dots$. The main lobe therefore extends from $-2\pi/\tau$ to $2\pi/\tau$ and its width is $4\pi/\tau$.
- The main-lobe width is inversely proportional to the pulse duration. A short pulse has a wide spectrum and a long pulse has a narrow spectrum. This inverse relationship between duration and bandwidth is a recurring theme in signal analysis.

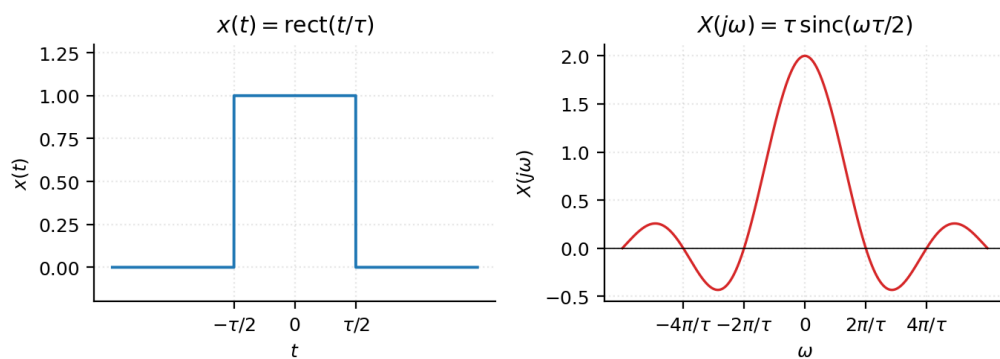


Figure 4: The rectangular pulse $\text{rect}(t/\tau)$ and its Fourier transform $\tau \text{sinc}(\omega\tau/2)$. The zero crossings are located at integer multiples of $2\pi/\tau$ and the main-lobe width is $4\pi/\tau$.

3.5 Fourier Series versus Fourier Transform

The relationship between the two representations becomes clear when a periodic signal is formed by repeating a pulse. Let $x(t)$ be a single pulse with transform $X(j\omega)$, and let $x_p(t)$ be the periodic signal obtained by repeating that pulse every T seconds. Comparing the analysis equations of the two representations gives

$$a_n = \frac{1}{T}X(jn\omega_0), \quad \omega_0 = \frac{2\pi}{T},$$

or equivalently $Ta_n = X(j\omega)|_{\omega=n\omega_0}$. In other words, the Fourier series coefficients of the periodic signal are samples of the Fourier transform of one period, scaled by $1/T$.

As T increases while the pulse itself is held fixed, the envelope $X(j\omega)$ does not change, but the spacing $\omega_0 = 2\pi/T$ between the samples decreases. In the limit $T \rightarrow \infty$ the samples merge into the continuous function $X(j\omega)$, which is exactly the limiting argument of Section 3.1. This is illustrated in Figure 5 for a periodic train of rectangular pulses, one period of which is the pulse of Section 3.4, so that the envelope is the sinc function obtained there. Note that this is a different signal from the bipolar square wave of Section 2.7, whose single period is not a rectangular pulse.

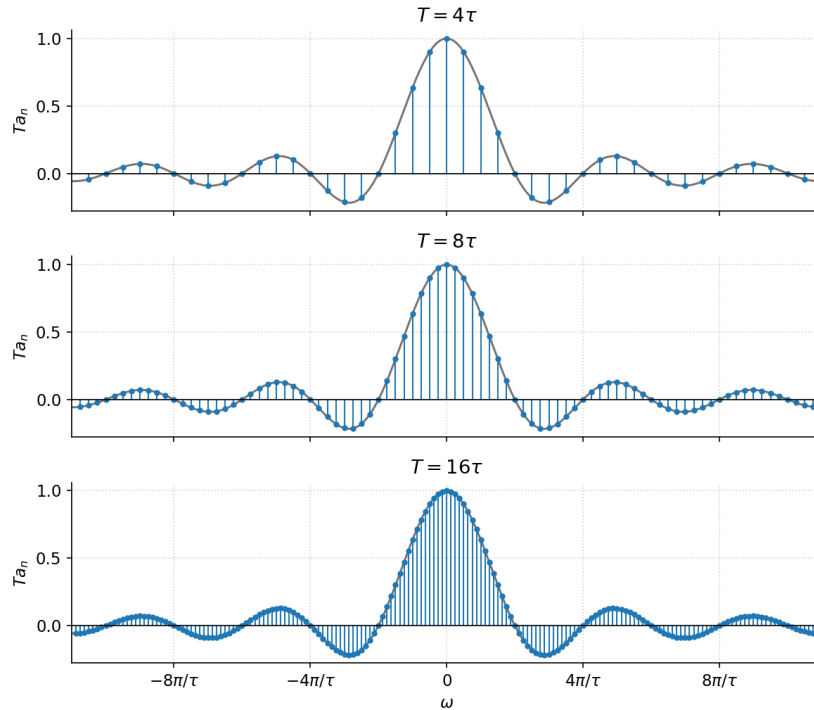


Figure 5: Fourier series coefficients (scaled as Ta_n) of a periodic pulse train for several values of the period T , with the pulse width τ held fixed. The grey curve is the Fourier transform of a single pulse. As T increases, the spectral lines become more closely spaced while the envelope remains unchanged; as $T \rightarrow \infty$ the lines form a continuum.

3.6 Worked Example: Transform of a One-Sided Exponential

Find the continuous-time Fourier transform of $x(t) = e^{-\alpha t}u(t)$, where $\alpha > 0$.

From the analysis equation,

$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt = \int_0^{\infty} e^{(-\alpha - j\omega)t} dt = \left. \frac{e^{(-j\omega - \alpha)t}}{-j\omega - \alpha} \right|_{t=0}^{\infty} = \frac{1}{\alpha + j\omega}.$$

The evaluation at the upper limit vanishes because $\alpha > 0$; if $\alpha \leq 0$ the signal is not absolutely integrable and the integral does not converge.

Expressing the result as $X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)}$ and using

$$|X(j\omega)| = \sqrt{(\operatorname{Re}\{X(j\omega)\})^2 + (\operatorname{Im}\{X(j\omega)\})^2}, \quad \angle X(j\omega) = \tan^{-1} \left(\frac{\operatorname{Im}\{X(j\omega)\}}{\operatorname{Re}\{X(j\omega)\}} \right),$$

together with

$$\frac{1}{\alpha + j\omega} = \frac{\alpha - j\omega}{\alpha^2 + \omega^2},$$

gives

$$|X(j\omega)| = \frac{1}{\sqrt{\alpha^2 + \omega^2}}, \quad \angle X(j\omega) = -\tan^{-1} \left(\frac{\omega}{\alpha} \right).$$

The magnitude spectrum is even and the phase spectrum is odd, which is again the conjugate symmetry expected of a real signal. Two reference points are commonly used:

$$\omega = 0 \Rightarrow |X(j\omega)| = \frac{1}{\alpha}, \quad \omega = \alpha \Rightarrow |X(j\omega)| = \frac{1}{\sqrt{2}\alpha}.$$

At $\omega = \alpha$ the magnitude has fallen to $1/\sqrt{2}$ of its peak value, which is the half-power or -3 dB point. The effective bandwidth of the signal is therefore proportional to α : the faster the exponential decays in time, the wider its spectrum. The phase decreases monotonically from $+\pi/2$ as $\omega \rightarrow -\infty$ to $-\pi/2$ as $\omega \rightarrow +\infty$, passing through zero at $\omega = 0$.

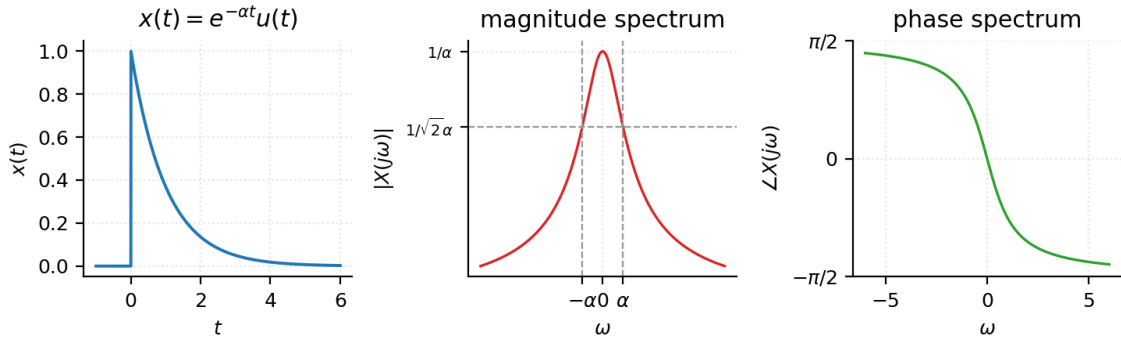


Figure 6: The one-sided exponential $x(t) = e^{-\alpha t}u(t)$ with $\alpha = 1$, together with its magnitude and phase spectra. The magnitude falls to $\frac{1}{\sqrt{2}\alpha}$ at $\omega = \pm\alpha$, and the phase varies from $+\pi/2$ to $-\pi/2$.

4 Properties of the Continuous-Time Fourier Transform

Throughout this section it is assumed that

$$x(t) \longleftrightarrow X(j\omega), \quad y(t) \longleftrightarrow Y(j\omega), \quad h(t) \longleftrightarrow H(j\omega).$$

The properties below are used constantly, because they make it possible to obtain new transforms from known ones without evaluating any integral.

4.1 Linearity

$$ax_1(t) + bx_2(t) \longleftrightarrow aX_1(j\omega) + bX_2(j\omega),$$

where a and b are arbitrary constants. This follows immediately from the linearity of the defining integral.

4.2 Conjugation and Symmetry

In general,

$$x^*(t) \longleftrightarrow X^*(-j\omega).$$

If $x(t)$ is a real-valued time signal, then

$$X(-j\omega) = X^*(j\omega),$$

where the asterisk denotes the complex conjugate. This single relation, called **conjugate symmetry**, has several consequences:

- $\text{Re}\{X(j\omega)\}$ is an even function of ω and $\text{Im}\{X(j\omega)\}$ is an odd function of ω .
- $|X(j\omega)|$ is even and $\angle X(j\omega)$ is odd.
- If $x(t)$ is real and even, then $X(j\omega)$ is real and even. The rectangular pulse of Section 3.4 is an example.
- If $x(t)$ is real and odd, then $X(j\omega)$ is purely imaginary and odd.
- For a real signal, the even part of $x(t)$ transforms to $\text{Re}\{X(j\omega)\}$ and the odd part transforms to $j \text{Im}\{X(j\omega)\}$.

A practical consequence is that for a real signal it is sufficient to plot the spectrum for $\omega \geq 0$.

4.3 Time Shifting and Frequency Shifting

$$x(t - t_0) \longleftrightarrow X(j\omega)e^{-j\omega t_0} \quad (\text{time shift}),$$

$$x(t)e^{j\omega_0 t} \longleftrightarrow X(j(\omega - \omega_0)) \quad (\text{frequency shift}).$$

A time shift does not change the magnitude spectrum; it adds the linear phase term $-\omega t_0$ to the phase spectrum. A frequency shift translates the whole spectrum along the frequency axis. Combining the frequency-shift property with Euler's relation gives the modulation result

$$x(t) \cos(\omega_0 t) \longleftrightarrow \frac{1}{2}X(j(\omega - \omega_0)) + \frac{1}{2}X(j(\omega + \omega_0)),$$

which is the basis of amplitude modulation in communication systems.

4.4 Time Scaling

Let α be a real constant, and consider $x(\alpha t)$. From the definition,

$$\mathcal{F}\{x(\alpha t)\} = \int_{t=-\infty}^{\infty} x(\alpha t)e^{-j\omega t} dt.$$

Substituting $\tau = \alpha t$, so that $dt = d\tau/\alpha$, and keeping track of the direction of integration when α is negative, gives

$$\mathcal{F}\{x(\alpha t)\} = \begin{cases} \frac{1}{\alpha} \int_{-\infty}^{\infty} x(\tau)e^{-j(\omega/\alpha)\tau} d\tau, & \alpha > 0, \\ -\frac{1}{\alpha} \int_{\infty}^{-\infty} x(\tau)e^{-j(\omega/\alpha)\tau} d\tau, & \alpha < 0. \end{cases}$$

The two cases are combined into the single statement

$$x(\alpha t) \longleftrightarrow \frac{1}{|\alpha|} X\left(\frac{j\omega}{\alpha}\right).$$

Compressing a signal in time therefore expands its spectrum in frequency, and expanding a signal in time compresses its spectrum, as illustrated in Figure 7. The special case $\alpha = -1$ gives the **time-reversal** property

$$x(-t) \longleftrightarrow X(-j\omega).$$

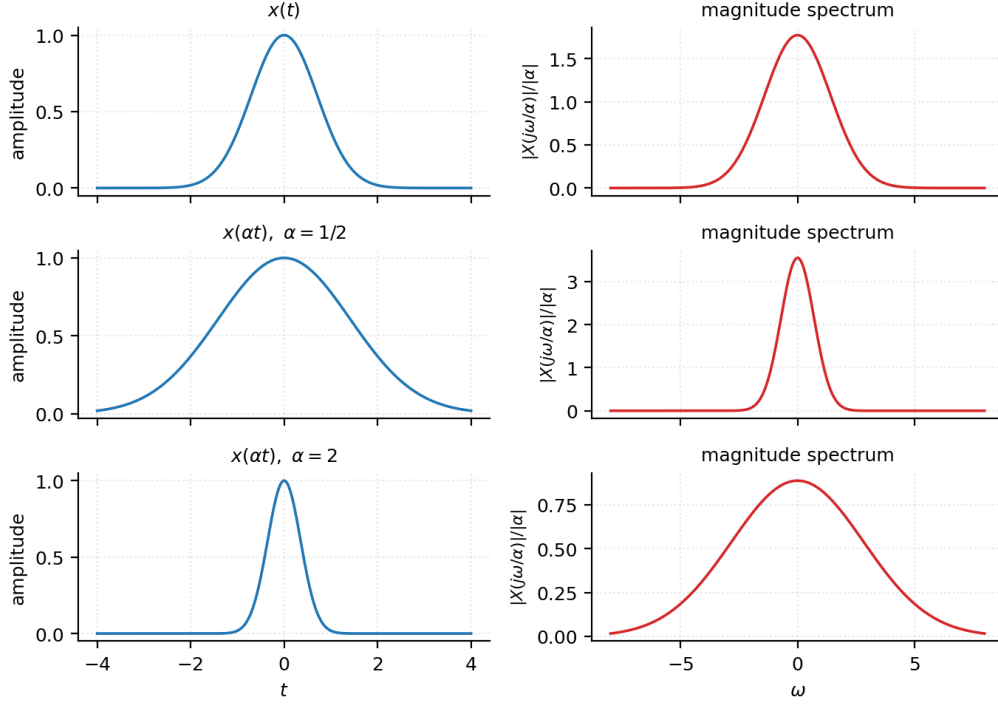


Figure 7: The time-scaling property. Top: the original signal and its magnitude spectrum. Middle: the time-expanded signal ($\alpha = 1/2$) and the resulting narrower spectrum. Bottom: the time-compressed signal ($\alpha = 2$) and the resulting wider spectrum.

4.5 Differentiation and Integration

Differentiating the synthesis equation with respect to t gives

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(j\omega), \quad \frac{d^n x(t)}{dt^n} \longleftrightarrow (j\omega)^n X(j\omega).$$

Differentiation therefore emphasises high frequencies. The corresponding integration property is

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(j0)\delta(\omega),$$

where the impulse term accounts for the average value, or DC content, that integration may introduce. Differentiating in the frequency domain instead gives

$$tx(t) \longleftrightarrow j \frac{d}{d\omega} X(j\omega).$$

4.6 Convolution

If $x(t) \leftrightarrow X(j\omega)$ and $h(t) \leftrightarrow H(j\omega)$, then

$$x(t) * h(t) \longleftrightarrow X(j\omega)H(j\omega).$$

To prove this statement, start from the definition of the transform of the convolution integral:

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \right] e^{-j\omega t} dt.$$

Interchanging the order of integration gives

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau) e^{-j\omega t} dt \right] d\tau.$$

From the time-shifting property, the bracketed term is simply $H(j\omega)e^{-j\omega\tau}$. Thus

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} x(\tau) H(j\omega) e^{-j\omega\tau} d\tau = H(j\omega) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau,$$

and therefore

$$\mathcal{F}\{x(t) * h(t)\} = H(j\omega) X(j\omega).$$

This is the single most useful property of the transform. For an LTI system with impulse response $h(t)$, the convolution in the time domain,

$$y(t) = x(t) * h(t),$$

becomes an ordinary multiplication in the frequency domain,

$$Y(j\omega) = X(j\omega) H(j\omega),$$

where $H(j\omega)$ is the frequency response of the system, consistent with the eigenfunction property of Section 1.3.

4.7 Multiplication

The dual of the convolution property states that multiplication in the time domain corresponds to convolution in the frequency domain:

$$x(t)y(t) \longleftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) Y(j(\omega - \theta)) d\theta = \frac{1}{2\pi} X(j\omega) * Y(j\omega).$$

The factor $1/2\pi$ should be noted; it does not appear in the convolution property.

4.8 Parseval's Relation

For an aperiodic signal of finite energy, the total energy may be computed in either domain:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega.$$

The quantity $|X(j\omega)|^2$ is called the **energy spectral density**, because it describes how the energy of the signal is distributed over frequency; the energy contained in the band $\omega_1 < |\omega| < \omega_2$ is obtained by integrating it over that band. The relation is the aperiodic counterpart of the average-power result of Section 2.10, and it shows that the transform preserves energy up to the factor $1/2\pi$ demanded by the convention adopted here.

4.9 Duality

It is often necessary to find the Fourier transform of a time signal that has the same form as an entry in the transform column of a table of Fourier transforms. Such a transform can be obtained by using the table backwards. To see how, write the inversion formula as

$$\int_{\omega=-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = 2\pi x(t).$$

Comparing this expression with the forward transform

$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

shows that there is a symmetry between the two equations: they are identical except for a sign change in the exponential, a factor of 2π , and the interchange of the variables involved. This symmetry leads to the **duality** property of the Fourier transform.

Replacing t with $-t$ gives

$$2\pi x(-t) = \int_{\omega=-\infty}^{\infty} X(j\omega) e^{-j\omega t} d\omega.$$

Since ω is only a dummy variable of integration, the names of the variables may now be exchanged, writing ω as t and t as ω :

$$2\pi x(-\omega) = \int_{t=-\infty}^{\infty} X(jt)e^{-j\omega t} dt.$$

The right-hand side is exactly the Fourier transform of the time function $X(jt)$, so that

$$X(jt) \longleftrightarrow 2\pi x(-\omega).$$

This statement is frequently written in the abbreviated form $X(t) \leftrightarrow 2\pi x(-\omega)$, in which $X(t)$ means the function $X(\cdot)$ evaluated with time as its argument. Note that no j appears in the argument of $x(\cdot)$, which is a function of a real variable. In practice the property says: if a known transform pair is read from right to left, an additional factor of 2π and a reflection of the independent variable are required.

4.10 Worked Example: Duality Applied to the Sampling Function

Consider the signal

$$x(t) = \text{sinc}\left(\frac{\omega_B t}{2}\right).$$

Find $X(j\omega)$.

Evaluating the defining integral directly is awkward. Instead, start from the pair established in Section 3.4,

$$g(t) = \text{rect}(t/\tau) \longleftrightarrow G(j\omega) = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right),$$

and apply duality, $G(jt) \leftrightarrow 2\pi g(-\omega)$:

$$\tau \text{sinc}\left(\frac{\tau t}{2}\right) \longleftrightarrow 2\pi \text{rect}\left(\frac{-\omega}{\tau}\right) = 2\pi \text{rect}\left(\frac{\omega}{\tau}\right),$$

where the last step uses the fact that the rectangular function is even. Setting $\tau = \omega_B$ and dividing both sides by ω_B gives the required result

$$\boxed{\text{sinc}\left(\frac{\omega_B t}{2}\right) \longleftrightarrow \frac{2\pi}{\omega_B} \text{rect}\left(\frac{\omega}{\omega_B}\right) = \begin{cases} \frac{2\pi}{\omega_B}, & |\omega| < \frac{\omega_B}{2}, \\ 0, & |\omega| > \frac{\omega_B}{2}. \end{cases}}$$

A sampling function in time therefore corresponds to a rectangular band in frequency, which is the ideal low-pass characteristic; conversely, a rectangular pulse in time corresponds to a sampling function in frequency. The two situations are compared in Figure 8.

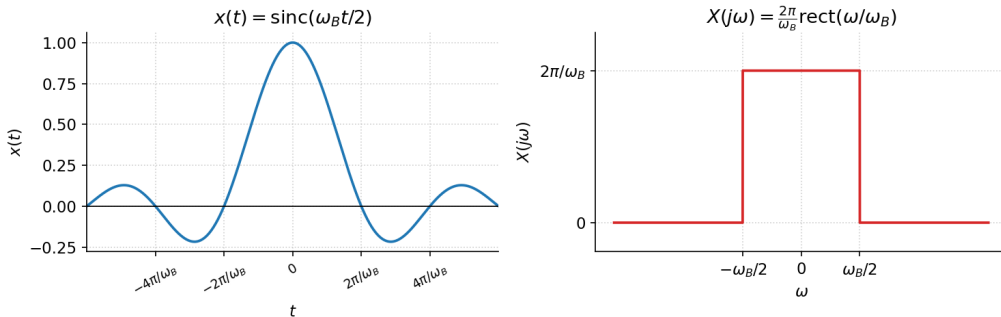


Figure 8: Duality illustrated: the sampling function $\text{sinc}(\omega_B t/2)$ in the time domain and its transform, a rectangular band of height $2\pi/\omega_B$ extending from $-\omega_B/2$ to $\omega_B/2$. Compare with Figure 4, in which the roles of the two domains are exchanged.

5 Fourier Transforms of Important Signals

5.1 The Unit Impulse

Applying the definition to $x(t) = \delta(t)$ and using the sampling property of the impulse gives

$$\mathcal{F}\{\delta(t)\} = \int_{t=-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = e^{-j\omega t} \Big|_{t=0} = e^0 = 1.$$

The impulse therefore contains all frequencies with equal weight and zero phase, which is why it is used as a test signal for measuring the frequency response of a system. Combining this result with the time-shifting property gives

$$\delta(t - t_0) \longleftrightarrow e^{-j\omega t_0}.$$

5.2 The Complex Exponential and Sinusoids

Consider the transform $X(j\omega) = 2\pi\delta(\omega - \omega_0)$ and ask what signal $x(t)$ it corresponds to. From the inverse transform,

$$x(t) = \frac{1}{2\pi} \int_{\omega=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0) e^{j\omega t} d\omega = e^{j\omega_0 t}.$$

Therefore

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0),$$

and with $\omega_0 = 0$ the constant signal gives $1 \leftrightarrow 2\pi\delta(\omega)$. Applying Euler's relations and linearity gives the transforms of the sinusoids:

$$\cos(\omega_0 t) \longleftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)], \quad \sin(\omega_0 t) \longleftrightarrow \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

None of these signals is absolutely integrable, so their transforms exist only in the generalised sense, that is, as impulses in the frequency domain.

5.3 Periodic Signals

Consider a periodic signal $x(t)$ with period T , so that $\omega_0 = 2\pi/T$, and assume that it has the Fourier-series representation

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}.$$

Taking the Fourier transform of both sides and using the transform of the complex exponential yields

$$X(j\omega) = \sum_{n=-\infty}^{\infty} a_n \mathcal{F}\{e^{jn\omega_0 t}\} = \sum_{n=-\infty}^{\infty} 2\pi a_n \delta(\omega - n\omega_0).$$

The Fourier transform of a periodic signal is therefore an impulse train in the frequency domain, with an impulse located at $\omega = n\omega_0$ of strength $2\pi a_n$, all impulses being separated from each other by ω_0 . The magnitude spectrum $|X(j\omega)|$ consists of impulses of strength $2\pi|a_n|$.

It should be noted that the Fourier transform of a periodic signal is generally not itself a periodic function of ω : although the impulses are equally spaced, their weights $2\pi a_n$ generally vary with n . The exception is the case in which the weights are themselves constant, as happens for the impulse train of Section 5.4, whose transform is periodic in ω . This result also reconciles the two representations: the line spectrum a_n and the impulsive transform $X(j\omega)$ carry exactly the same information.

5.4 The Impulse Train

Consider the periodic signal

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$

To find its transform, first compute the Fourier series coefficients. Over one period only the impulse at the origin lies inside the interval of integration, so

$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn(2\pi/T)t} dt = \frac{1}{T}.$$

The series representation is therefore

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(2\pi/T)t} = \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn(2\pi/T)t},$$

and applying the result of the previous section gives

$$X(j\omega) = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi n}{T}\right).$$

The Fourier transform of a sequence of impulses in the time domain is thus a sequence of impulses in the frequency domain. The spacing in time, T , and the spacing in frequency, $2\pi/T$, are inversely related: closely spaced impulses in time correspond to widely spaced impulses in frequency. This transform pair is the starting point for the analysis of sampling.

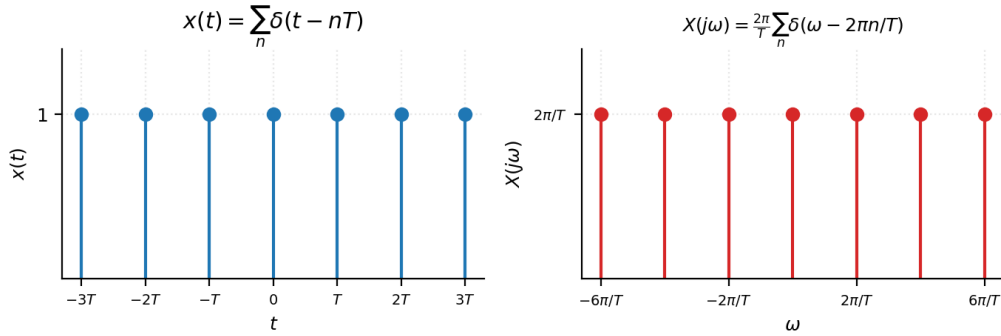


Figure 9: The impulse train in the time domain and its Fourier transform. The time-domain impulses have unit strength and spacing T ; the frequency-domain impulses have strength $2\pi/T$ and spacing $2\pi/T$.

5.5 The Unit Step

The unit step is not absolutely integrable, but its transform follows from the integration property. Since

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau \quad \text{and} \quad \mathcal{F}\{\delta(t)\} = 1,$$

the integration property with $X(j0) = 1$ gives

$$u(t) \longleftrightarrow \frac{1}{j\omega} + \pi\delta(\omega).$$

The term $1/(j\omega)$ describes the transition at $t = 0$ and the impulse $\pi\delta(\omega)$ accounts for the non-zero average value of the step.

6 Summary Tables

Table 3 collects the properties of the continuous-time Fourier transform, and Table 4 lists the basic transform pairs derived in this lecture. Both are reproduced in the same form in the reference texts listed in Section 7.

Property	Signal	Fourier transform
	$x(t)$	$X(j\omega)$
	$y(t)$	$Y(j\omega)$
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Time shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
Frequency shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
Conjugation	$x^*(t)$	$X^*(-j\omega)$
Time reversal	$x(-t)$	$X(-j\omega)$
Time and frequency scaling	$x(\alpha t)$	$\frac{1}{ \alpha } X\left(\frac{j\omega}{\alpha}\right)$
Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta)Y(j(\omega - \theta)) d\theta$
Differentiation in time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{j\omega} X(j\omega) + \pi X(j0)\delta(\omega)$
Differentiation in frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
Modulation	$x(t) \cos(\omega_0 t)$	$\frac{1}{2} X(j(\omega - \omega_0)) + \frac{1}{2} X(j(\omega + \omega_0))$
Duality	$X(jt)$	$2\pi x(-\omega)$
Parseval	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$	
Conjugate symmetry	$x(t)$ real	$X(-j\omega) = X^*(j\omega)$
Symmetry	$x(t)$ real and even	$X(j\omega)$ real and even
Symmetry	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd

Table 3: Properties of the continuous-time Fourier transform.

7 References

- Samir S. Soliman and Mandyam D. Srinath, **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998). Chapter 3, Fourier series (Sections 3.3, 3.4); Chapter 4, Fourier transform (Sections 4.2, 4.3).
- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab, **Signals and Systems**, 2nd Edition, Prentice Hall (1996). Chapters 3 and 4.
- Richard G. Lyons, **Understanding Digital Signal Processing**, 3rd Edition, Prentice Hall (2010). Chapter 3, the discrete Fourier transform. (The 1st and 2nd editions appeared in 1996 and 2004; there is no 2001 edition.)

Signal	Fourier transform
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
$x(t) = 1$	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$\frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$e^{-\alpha t}u(t), \text{ Re}\{\alpha\} > 0$	$\frac{1}{\alpha + j\omega}$
$te^{-\alpha t}u(t), \text{ Re}\{\alpha\} > 0$	$\frac{1}{(\alpha + j\omega)^2}$
$\text{rect}(t/\tau)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$
$\text{sinc}\left(\frac{\omega_B t}{2}\right)$	$\frac{2\pi}{\omega_B} \text{rect}\left(\frac{\omega}{\omega_B}\right)$
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi n}{T}\right)$
$\sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$	$2\pi \sum_{n=-\infty}^{\infty} a_n \delta(\omega - n\omega_0)$

Table 4: Basic continuous-time Fourier transform pairs, using the convention $\text{sinc}(x) = \sin(x)/x$.